

Multiple Choice

1. **Answer:** C

Options: A) Following up with those who did not respond to the survey the first time; B) Asking questions in a neutral manner to avoid influencing the responses; C) Using stratified random sampling rather than simple random sampling; D) Selecting samples randomly

Note: Correct option: C - Using stratified random sampling rather than simple random sampling

2. **Answer:** B

Options: A) 5; B) 10; C) 15; D) 20

Note: Correct option: B - 10

3. **Answer:** A

Options: A) 7; B) 5; C) 2; D) 25

Note: Correct option: A - 7

4. **Answer:** C

Options: A) 4.9 miles; B) 5.8 miles; C) 6.0 miles; D) 6.3 miles

Note: Correct option: C - 6.0 miles

5. **Answer:** A

Options: A) $\frac{1}{2}$; B) $\frac{1}{4}$; C) $\frac{1}{8}$; D) $\frac{3}{5}$

Note: Correct option: A - $\frac{1}{2}$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '314'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '30 minutes'.

Long Form Response

11. **Answer:** A customer makes one of two choices for each condiment, to include it or not to include it. The choices are made independently, so there are $2^8 = 256$ possible combinations of condiments. For each of those combinations there are three choices regarding the number of meat patties, so there are altogether $(3)(256) = 768$ different kinds of hamburgers.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** p-3. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.